A 7B instruct model fails all 60 native Taglish prompts: English drift and unnatural code-switching

PinoAI research note · FINDING-001

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Abstract

We ran Qwen2.5-7B-Instruct on the 60 native-authored Taglish prompts of PH-EVAL v0.1. They cover workplace, casual, technical and academic registers, 15 each, all written and verified by a native speaker. The model passed 0 of 60 (95% Wilson interval 0–6.0%). The failures have two forms:

- **English drift.** Our LLM judge (a commercial closed-API model, used zero-shot) classifies 20 of 60 answers as English, 16 of them with confidence ≥ 0.7.
- **Unnatural code-switching.** Every answer the native reviewer rated for naturalness scored 0 or 1 on a 0–3 scale (n = 35).

The result holds when the 24 answers cut off at 512 tokens are excluded (0/36).

The same data is calibration #1 of the LLM judge. When confident, the judge agreed with the native reviewer on 14 of 20 items (Cohen's $\kappa = 0.37$). Its errors were mostly false passes (5 vs 1). Its FAIL verdicts are therefore usable, but PASS verdicts need human review.

This is an exploratory finding on one model. It motivates, but does not test, RESEARCH_THESIS H2 and H5.

1. Introduction

Code-switching between Filipino and English is how many Filipinos actually write at work, in school and online. Existing Filipino benchmarks are mostly monolingual Tagalog. Their Taglish coverage is limited to classification (sentiment and toxicity) [1], [2], so no public benchmark checks whether a model can *answer* in natural Taglish. PH-EVAL v0.1 added 60 native Taglish prompts for exactly that. This note reports the first model run on them, and what it says about our automatic judge.

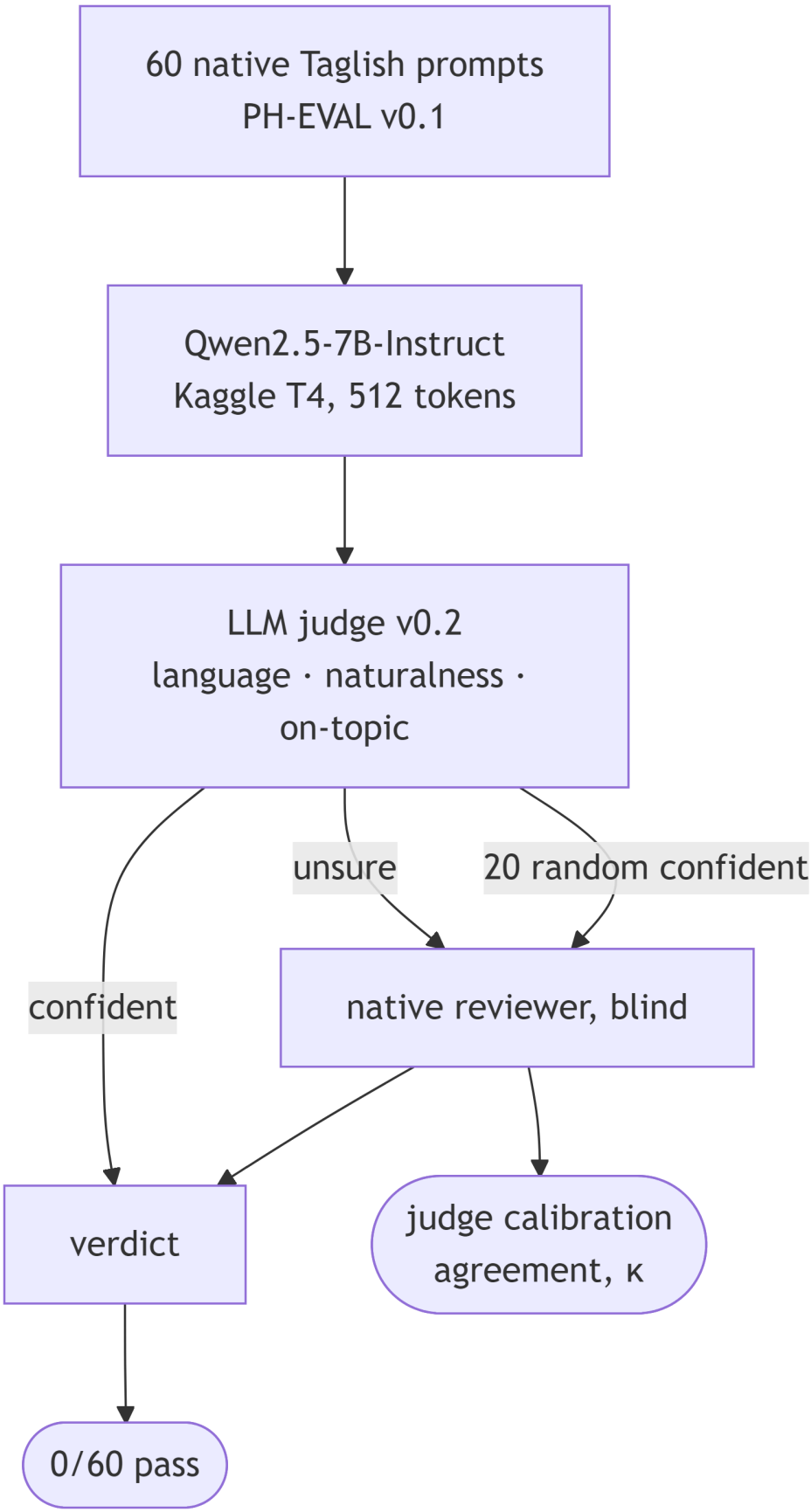
2. Pre-registered hypothesis

None: this is exploratory. The run was a calibration run for our LLM judge, not a pre-registered test. It relates to two hypotheses in our research thesis: H2 (explicit Taglish training yields more natural code-switching) and H5 (a Philippine-specific suite exposes failures that standard benchmarks miss). It can motivate them, but not test them: H2 needs a trained model, and H5 needs the same model's standard-benchmark scores for contrast.

3. Method

- **Model:** `Qwen/Qwen2.5-7B-Instruct` rev. a09a35458c702b33eeacc393d103063234e8bc28, fp16, transformers 5.17.0, max_new_tokens 512 (run_meta.json). Run on a single NVIDIA Tesla T4 (Kaggle), 2026-09-29. GPU hours for this run were not logged.
- **Items:** `data/eval/ph_eval/taglish/{workplace,casual,technical,academic}.jsonl`, 60 items, native-authored and human_verified, eval_type free_generation.
- **Judge:** judge policy v0.2, which asks bounded questions: answer language (Taglish/Filipino/English/Other), Taglish naturalness (0–3), on-topic. A verdict counts only when confident; otherwise the item is escalated to a human.
- **Human review:** The project's native-speaker reviewer, who also wrote the prompts, rated 62 items blind (the judge's verdict hidden): all 42 escalations, plus 20 random confident verdicts (seed 42, truncated answers excluded). `human_ratings.jsonl`, `calibration_key.json`.

- **Final verdict per item:** the judge's confident verdict, else the reviewer's rating.
- **Statistics:** 95% Wilson intervals for proportions, and Cohen's κ for judge–human agreement. Everything is recomputed by `experiments/reports/_figures.py`.



4. Results

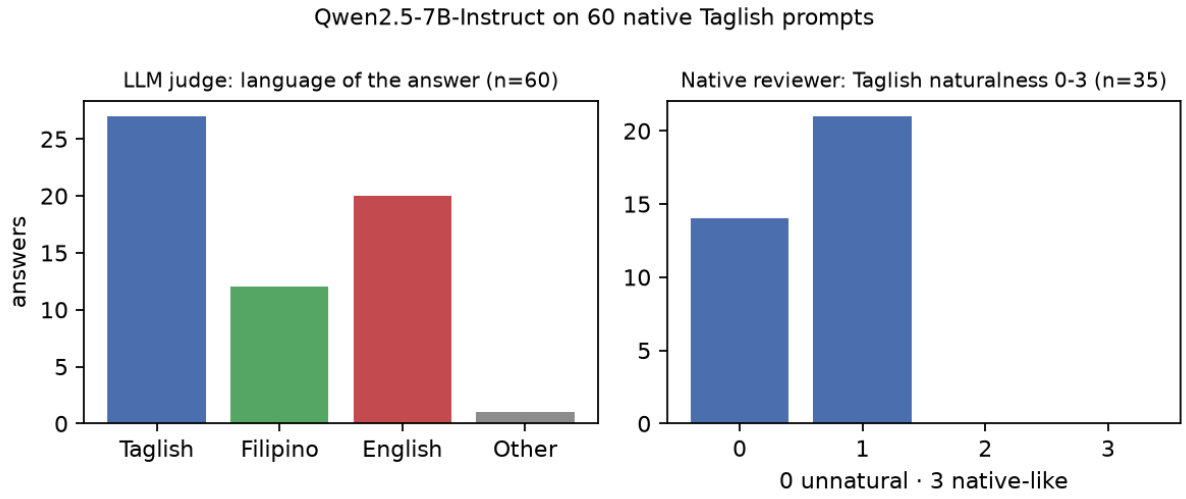


Figure 1: Left: the language of each answer, per the LLM judge (argmax). Right: the native reviewer's naturalness ratings on the Taglish items they rated. No answer reached 2 ("natural") or 3 ("native-like").

Table 1: Taglish outcome for Qwen2.5-7B-Instruct. Source: judge verdicts, human_ratings.jsonl, truncated_ids.json.

measure	value
Taglish items	60
final PASS (confident judge verdict, else reviewer)	0/60
95% Wilson CI for the pass rate	0.000 to 0.060
decided by: confident FAIL (judge)	34
decided by: escalated ⟶ reviewer	26
judge language choice = English (any confidence)	20
judge language choice = English, confidence ≥ 0.7	16
judge language choice = Taglish / Filipino / Other	27 / 12 / 1
PASS in academic	0/15
PASS in casual	0/15
PASS in technical	0/15
PASS in workplace	0/15
judge on_topic p ≥ 0.85 / p ≤ 0.15 (confident on / off topic)	40 / 0
answers cut at 512 tokens (truncated_ids.json)	24/60
PASS among NOT truncated answers	0/36 (Wilson 95% 0.000 to 0.096)

Table 2: LLM judge calibration #1 against the native reviewer. Source: calibration_key.json and human_ratings.jsonl.

measure	value
confident sample (random, seed 42, truncated excluded)	20
agreement judge vs native reviewer	14/20 (0.48 to 0.85 Wilson 95%)
Cohen's kappa	0.37
false PASS (judge pass, reviewer fail) / false FAIL	5 / 1
escalated items resolved by reviewer: fail / pass	39 / 3
judge Taglish score (0-1) where reviewer rated 0	mean 0.41 (n=14)
judge Taglish score (0-1) where reviewer rated 1	mean 0.64 (n=21)

- **The failure is complete and holds in every register.** 0/15 in each of workplace, casual, technical and academic.
- **Truncation does not explain it.** 0/36 among the answers that were not cut off.
- **Where the judge and the reviewer both judged a Taglish item, they agreed:** 9 of 9 in the confident sample.
- **The judge's naturalness score ranks answers the same way the reviewer does:** mean 0.41 where the reviewer gave 0, and 0.64 where they gave 1. The top of the scale (2–3) is untested, because no answer reached it.

5. Discussion

The model does not seem to fail because it misunderstands Taglish. The judge is confident that 40 of 60 answers are on-topic and that none is off-topic (Table 1), yet the model answered in English or stiff Taglish. Two practical points follow for PinoAI:

1. **Answering in Taglish is a separate skill from understanding it.** A model can read "Pa-check naman ng report bago mag-lunch" and still reply in formal English. Instruction data with natural Taglish *responses* is the likely lever. H2 predicts this; our SFT phase can test it.
2. **Automatic judging is safe in one direction only.** At $\kappa = 0.37$ with mostly false passes, the judge would inflate scores if we trusted its PASS verdicts. The interim policy (judge FAIL accepted; PASS on privacy, hallucination and instruction following goes to a human) follows from Table 2. It needs revalidation on fresh outputs (judge policy v0.3 is fitted to this very set, so it cannot be scored here).

> ℹ Correction (2026-10-02)
>
> The illustrative Taglish example in point 1 was replaced, because the earlier example matched an evaluation item.

6. Limitations and threats to validity

- **One model, one run.** Qwen2.5-7B-Instruct is not one of our five DEC-001 candidates. Nothing here ranks them.
- **One native reviewer.** There is no inter-annotator agreement yet. The reviewer also wrote the prompts, which is a possible familiarity bias.
- **Small n.** With 60 items, a true pass rate as high as 6.0% cannot be ruled out (Wilson upper bound).
- **Truncation.** 24 of 60 answers hit the 512-token cap; handled by the 0/36 check.
- **Language labels come from the judge**, which is weakest exactly on Taglish. The 16 confident "English" labels are the reliable subset.
- **No pre-registration.** Treat this as hypothesis-generating.

7. Next steps

- Run the same 60 prompts on the five candidates' instruct siblings, for DEC-001 Part A and a multi-model version of this note.
- Add a second native reviewer on a sample, to get inter-annotator agreement (needed for Paper 1).
- Validate judge policy v0.3 on those fresh outputs before relaxing the PASS escalation policy.
- Pre-register H2 as a concrete SFT experiment (Taglish-response data vs translated data), once the SFT phase opens.

8. Update, 2026-09-30: newer models do not show the English drift

After this note, three newer instruct models were run on the same 60 Taglish prompts, among the 175 PH-EVAL items, under the same settings (see the RESULT-001 note):

Table 3: Sources: function-word share from the project's rule-based language-share metric; naturalness means from the single Part A computation (`experiments/results/DEC-001-partA.txt`).

Model	Median Tagalog function-word share (Taglish items)	Human naturalness, mean 0–3 (n)
Qwen2.5-7B-Instruct (this note)	not computed on this metric	≤ 1 on every rated answer (35)
Gemma 4 12B	0.97	2.30 (56)
Qwen3.5-9B	0.99	1.96 (57)
Sailor2-8B	0.97	1.10 (49)

Read the update this way:

- **English drift appears specific to the older model.** None of the three newer models drifts to English.
- **Naturalness varies widely among newer models:** from 1.10 to 2.30 on the 0–3 scale.
- **Our metric catches drift, not quality.** The function-word share detects drift to English, but not naturalness: natural Taglish keeps Tagalog function words and switches content words.

So this note's title result (0/60 for Qwen2.5-7B-Instruct) stands. But its implication narrows: the gap for PinoAI to close is *natural* Taglish, not merely *answering in Taglish at all*.

The second-sheet ratings behind these means carry the blinding caveat reported in RESULT-001.

Acknowledgments

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AI assistance. Analysis code, statistical checks and drafts of this document were prepared with the help of Claude (Anthropic), an AI model. Every number is generated from the project's result files. The author reviewed the content and takes responsibility for it.

References

[1] J. R. Montalan and others, "Batayan: A Filipino NLP benchmark for evaluating Large Language Models," *arXiv preprint arXiv:2502.14911*, 2025.

[2] L. J. V. Miranda and others, "FilBench: Can LLMs Understand and Generate Filipino?," *arXiv preprint arXiv:2508.03523*, 2025.